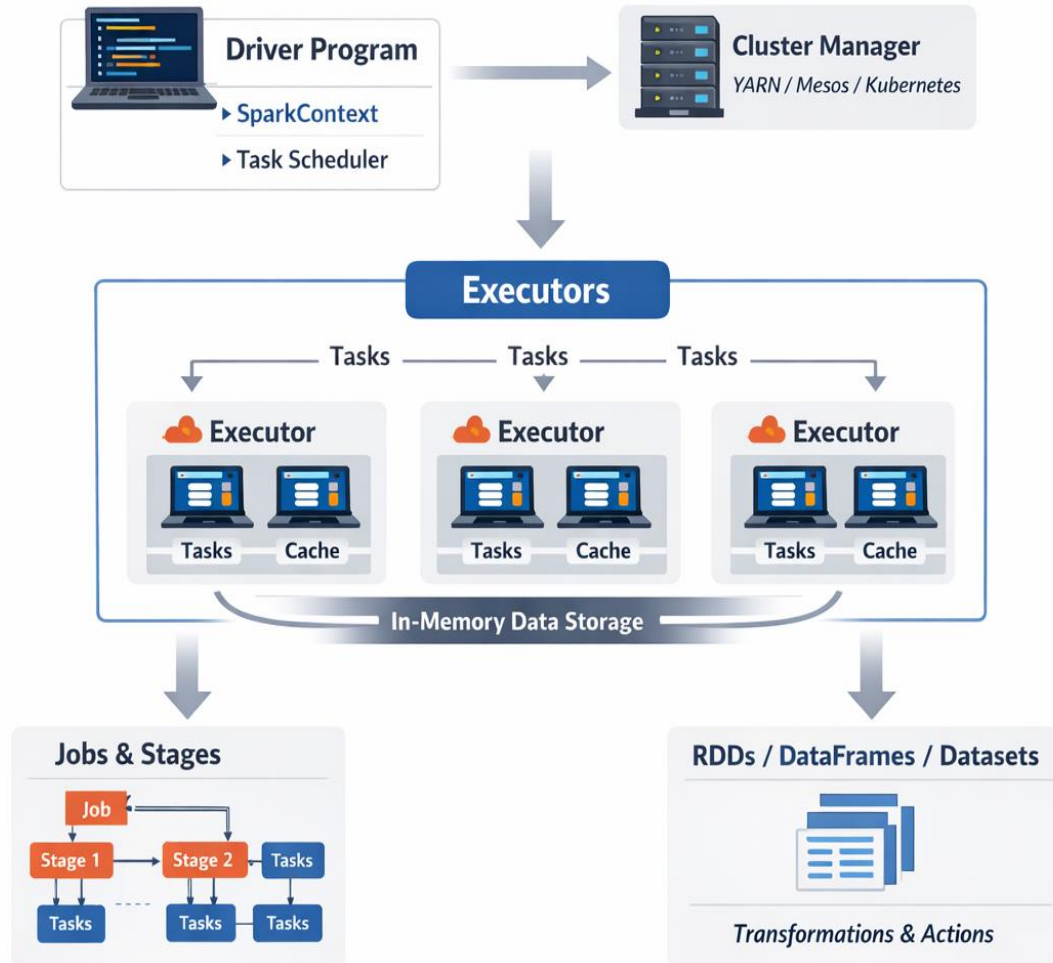


# Databricks Spark Architecture

Beginner-friendly overview  
Apache Spark on Databricks

# ⚡ Apache Spark Architecture 🚀



# What is Spark on Databricks?

- Databricks is built on Apache Spark
  - Provides optimized Spark runtime
  - Used for large-scale data processing
  - Supports batch, streaming, ML & analytics

# Spark Cluster Overview

- Driver Node – controls execution
  - Worker Nodes – execute tasks
  - Executors – perform actual processing
  - Cluster managed automatically by Databricks

# Driver Node

- Main brain of Spark application
  - Converts code into DAG
  - Schedules tasks to executors
  - Tracks job execution & failures

# Worker Nodes & Executors

- Worker nodes host executors
  - Executors run tasks in parallel
  - Each executor has CPU & memory
  - Caches data for faster processing

# Databricks Cluster Manager

- Allocates resources dynamically
  - Auto-scaling clusters
  - Auto-termination to save cost
  - Handles fault tolerance

# Storage Layer

- Uses ADLS Gen2 / S3 / GCS
  - Delta Lake for ACID transactions
  - Supports batch & streaming data
  - Schema enforcement & time travel



# Spark Execution Flow

- User submits code via notebook
  - Driver creates DAG & execution plan
  - Tasks sent to executors
  - Results written to Delta / Data Lake

# Why Spark on Databricks?

- High performance & scalability
  - Optimized Spark runtime (Photon)
  - Easy collaboration with notebooks
  - Enterprise-grade security & governance

# Thank You

- Understanding Spark architecture is key
  - Essential skill for Data Engineers
  - Happy Learning & Happy Connecting!